ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 2 SET C **SUMMER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 20**

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

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1. Use Karnaugh map to find the minimal form for the following expression. Clearly group the terms contributing to your solution. 5 (CO3) (PO2)

$$F(A,B,C,D,E) = \sum m(0,1,4,5,6,8,13,16,21,22,26,29) + \sum d(7,17,20,25)$$

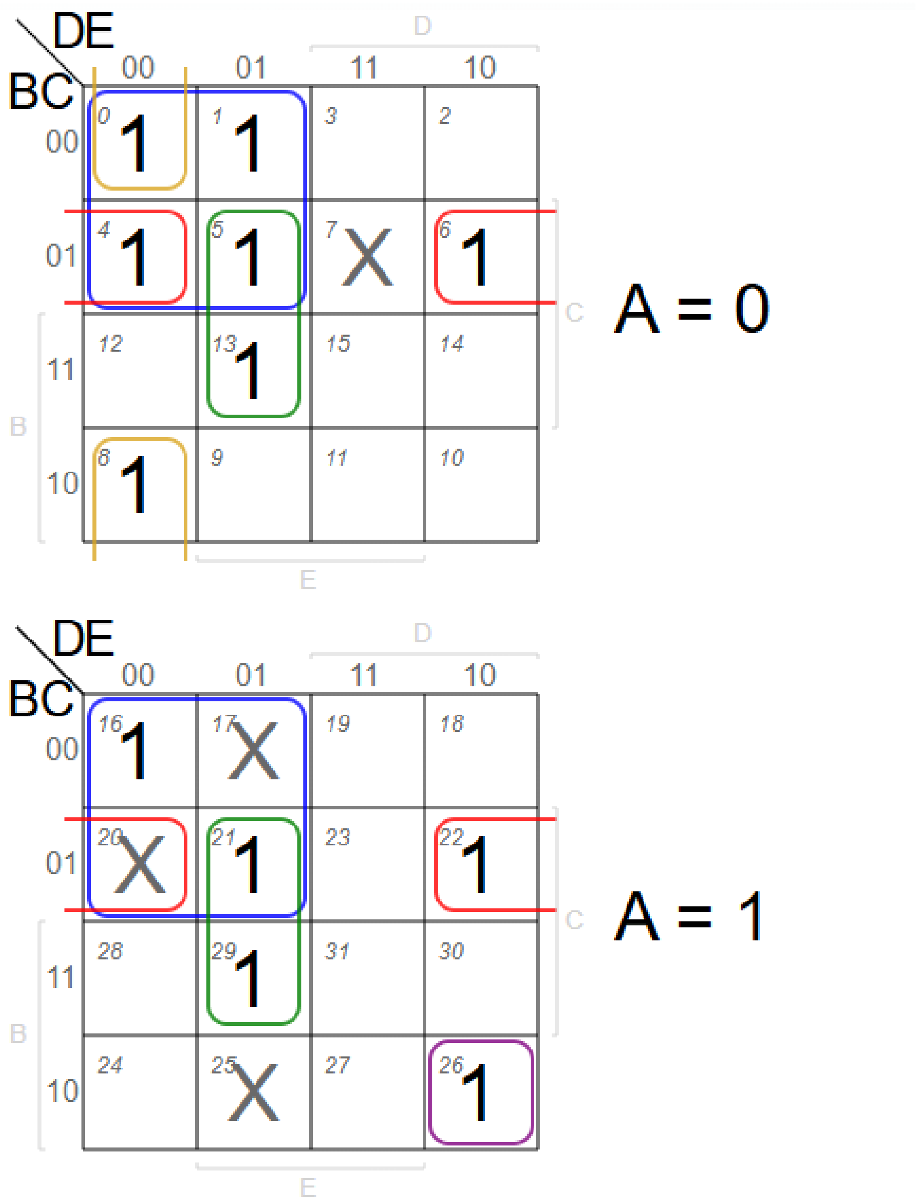
Solution:

Final Answer:

$$\boxed{F = B'D' + B'CE' + CD'E + A'C'D'E' + ABC'DE'}$$

Rubrics:

- 0.5 point each for correctly grouping each term (times 5).
- 0.5 point each for correctly identifying each term (times 5).
- No point for drawing the K-map since its already given.



F = B'D' + B'CE' + CD'E + A'C'D'E' + ABC'DE'

Figure 1: K-map Solution

2. Simplify the following expression using Quine McCluskey (Tabular Method). And in the process identify the Prime Implicants and Essential Prime Implicants. 11 (CO5) (PO3)

$$F(A,B,C,D) = \sum m(0,6,7,12,14) + \sum d(9,15)$$

Solution:

For detailed solution visit this site. **Final Answer:**

$$\boxed{F = A'B'C'D' + ABD' + BC}$$

Rubrics:

- 1 point for correctly writing the binary representation of each minterm.
- 1 point for correctly grouping minterms based on number of 1s in their binary representation.
- 3 points for correct matching table 1.
- 3 points for correct matching table 2.
- 2 points for correct Prime Implicant Table.
- 1 point for correctly identifying Essential Prime Implicants.

3. Simplify the following Boolean expression using the Cut and Try method, and draw the simplified Boolean circuit using only NAND gates. 4 (CO2) (PO2)

$$F = ab'c + a'c + ab$$

Solution:

Given,

$$F = a\overline{b}c + \overline{a}c + ab$$

$$\Rightarrow F = a\overline{b}c + ab + \overline{a}c$$

$$\Rightarrow F = a(\overline{b}c + b) + \overline{a}c$$

$$\Rightarrow F = a(\overline{b} + b)(c + b) + \overline{a}c$$

$$\Rightarrow F = a(c + b) + \overline{a}c$$

$$\Rightarrow F = ac + ab + \overline{a}c$$

$$\Rightarrow F = c(a + \overline{a}) + ab$$

$$F = ab + c$$

NAND Representation -

$$F = \overline{\overline{c}.\overline{ab}}$$

Try to draw it on your own.

Final Answer:

$$\boxed{F = ab + c}$$

$$\boxed{F = \overline{\overline{c}.\overline{ab}}}$$

Rubrics:

- 2 points for correct simplification.
- 2 points for correct NOR representation.